TRAVEL INSURANCE PURCHASE PREDICTION

Data Science & Machine Learning Project Report

1. Introduction

This project develops a machine learning system to predict whether a customer is likely to purchase travel insurance. It uses demographic, financial, health-related and travel-behaviour features and compares multiple classification algorithms.

2. Objectives

• Clean and preprocess the dataset.

• Perform exploratory data analysis and visualization.

• Create an income-per-family-member feature.

• Compare Logistic Regression, Decision Tree and Random Forest.

• Evaluate Accuracy, Precision, Recall, F1 Score and ROC-AUC.

• Identify important features and demonstrate a sample prediction.

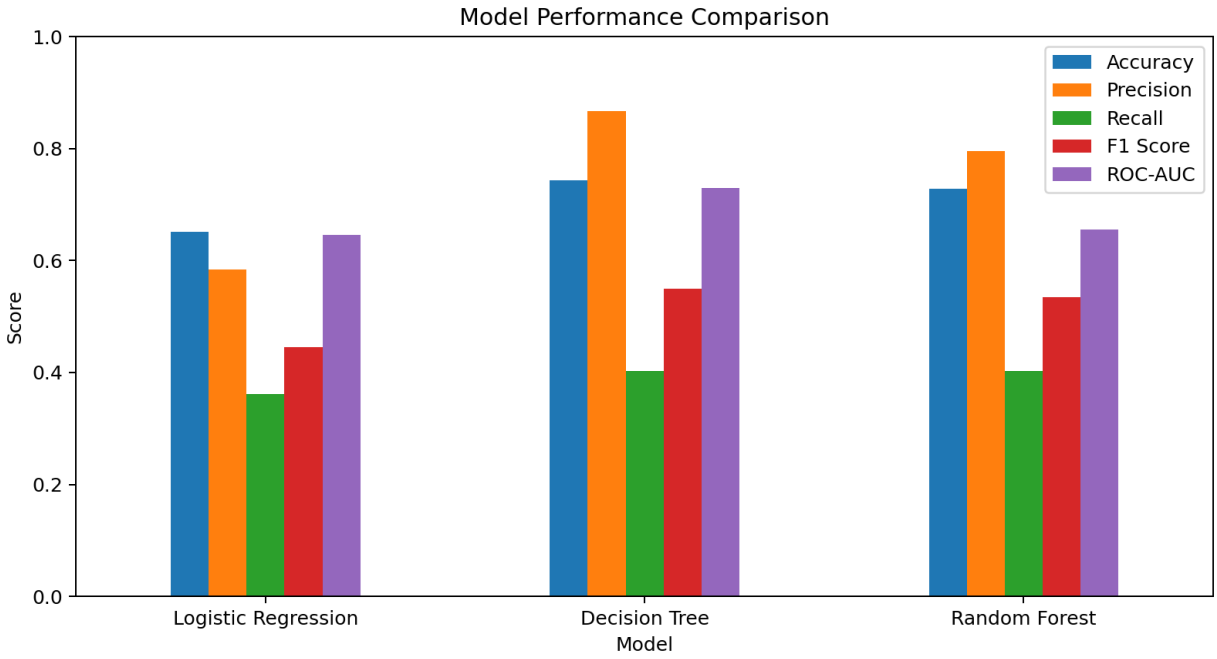
3. Dataset and Preprocessing

The supplied dataset contained 1,987 rows and 10 columns before cleaning. The target is TravelInsurance (0 = no purchase, 1 = purchase). The Unnamed: 0 index column was removed. The duplicate check found 738 exact duplicate rows, leaving 1,249 unique records.

No missing-value imputation was required in the supplied dataset. Categorical variables were one-hot encoded, numerical variables were standardized, and the data was split 80/20 with stratification.

4. Exploratory Data Analysis

The analysis included distributions for age and annual income, insurance purchase counts, categorical comparisons, box plots and a correlation heatmap.



5. Machine Learning Methodology

Three classification models were tested: Logistic Regression, Decision Tree and Random Forest. The Decision Tree was selected for the final sample-prediction demonstration because it produced the highest F1 Score in this run. F1 Score balances precision and recall.

6. Model Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.6520	0.5833	0.3608	0.4459	0.6459
Decision Tree	0.7440	0.8667	0.4021	0.5493	0.7294
Random Forest	0.7280	0.7959	0.4021	0.5342	0.6552

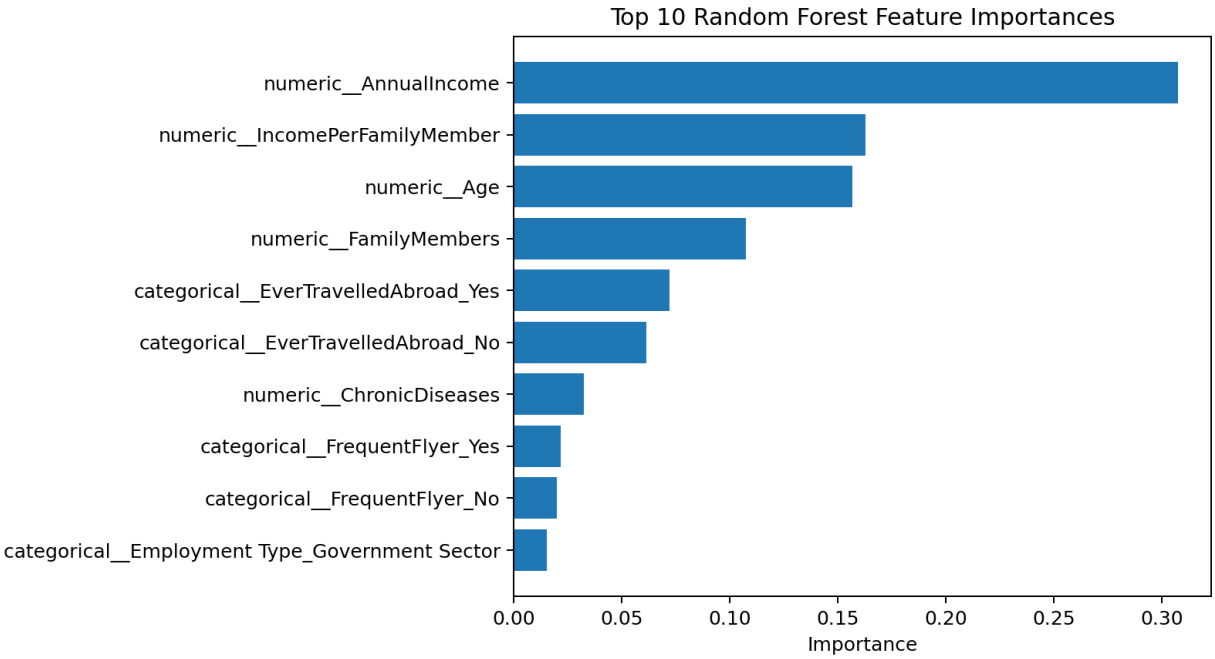
Selected model: Decision Tree. Test-set results: Accuracy 0.7440, Precision 0.8667, Recall 0.4021, F1 Score 0.5493, ROC-AUC 0.7294.

7. Confusion Matrix

The selected Decision Tree produced the test-set confusion matrix [[147, 6], [58, 39]], with rows representing actual classes and columns representing predicted classes.

8. Feature Importance

Random Forest feature importance identified AnnualIncome, IncomePerFamilyMember, Age and FamilyMembers among the largest importance values, followed by previous overseas travel indicators.



9. Key Insights

• The cleaned dataset contains 1,249 unique records.

• AnnualIncome had the largest Random Forest feature-importance value in this run.

• IncomePerFamilyMember, Age and FamilyMembers were also among the most important features.

• Previous overseas travel and frequent-flyer indicators added predictive information.

• Different models showed different precision/recall trade-offs, so multiple metrics were reported.

10. Challenges and Future Improvements

• The supplied dataset contained many exact duplicate records, which were removed.

• Additional customer, trip and policy-related variables could improve the model.

• Cross-validation and hyperparameter tuning could provide more robust performance estimates.

• Testing on an independent dataset would help assess generalization.

11. Conclusion

The project demonstrates a complete workflow for travel insurance purchase prediction, from data cleaning and EDA through feature engineering, classification, evaluation and interpretation. The results provide a baseline model and identify several customer characteristics that were informative in this dataset.